Agriculture and Ecology: A Domain in Which Three Kinds of Edge Converge

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1. Purpose of the Investigation in This Domain

Figure 1: Agriculture and Ecology: A Domain in Which Three Kinds of Edge Converge —

This report presents the results of an investigation into the structural correspondences between the major theoretical and practical frameworks of agriculture and ecology and the five-stage model of creation, namely Field ( ) -> Wave ( ) -> Edge ( ) -> Vortex ( ) -> Bundle ( ).

Agriculture and ecology are domains in which cycles of disturbance and recovery, management and reevaluation, are presupposed. How do ecosystems change, stabilize, and then change again? How do human beings intervene in that process? The purpose of this investigation is to examine how the five-stage model corresponds to this cyclical structure.

This report is a presentation of investigative findings. It is neither an introductory text on agriculture nor a proof document for the five-stage model. It reports both the places where correspondence was found and the places where correspondence is weak.

2. Method of Investigation

In this investigation, 10 theoretical and practical frameworks from the domain of agriculture and ecology were selected as candidates, and their structural correspondence with the five-stage model was evaluated.

Selection criteria: The investigation covered a broad range, from foundational ecological theories such as succession, regime shifts, and self-organization to agricultural management practices such as IPM, nitrogen management, and rangeland management. Particular emphasis was placed on theories concerning boundaries, thresholds, and feedback.

Evaluation method: For each candidate, three evaluative lenses were established. First, what type of Edge is implemented in the framework: threshold type, boundary type, or relational type. Second, whether feedback, that is, Vortex, is explicitly present. Third, whether there is a connection to agricultural management intervention. The resulting assessment was cross-checked through independent review.

Criteria of judgment: A candidate was judged to show "strong correspondence" when correspondence could be confirmed for each of the five stages and when at least two of Edge's three conditions, namely relational network, indeterminacy, and connection to Vortex, were present at high intensity.

3. Overview of the Five-Stage Model

The five-stage model used in this investigation is outlined below.

Field: An undifferentiated state. It corresponds to initial conditions in which neither direction nor structure has yet been determined.

Wave: A stage of exploratory divergence in which multiple directions spread and compete. Differences become manifest, and fluctuation appears.

Edge: A state of tension in which opposing elements coexist without converging on either side. Multiple relations intersect, and the situation cannot be resolved by a simple either-or choice. Whether one can remain in this stage is deeply tied to the quality of creation.

Vortex: A stage in which a new coherence, or order, rises spontaneously out of tension. A structure that did not previously exist emerges.

Bundle: A stage in which form becomes fixed and stabilizes as a reusable structure. Bundle becomes the precondition for the next Field, and the cycle continues.

If these definitions produce a sense of incongruity, that itself is an important perspective for the investigation. There is meaning both in continuing to read while retaining that sense of incongruity and in pausing here once before proceeding.

4. Results of the Investigation

Overall Picture

All 10 candidates evaluated from the domain of agriculture and ecology were judged adoptable under the criteria of this investigation. In terms of the strength of correspondence with Edge, 5 received the highest evaluation and 5 received a moderate evaluation.

The most distinctive feature of this domain is that Edge appears in three different forms. There were 4 instances of the threshold type, in which Edge appears as a bifurcation point; 3 instances of the boundary type, in which Edge appears as an interface; 2 instances of the relational type, in which Edge appears as a network; and 1 composite instance. It is rare, even among the 30 domains, to find within a single domain such a diverse set of implementations of Edge.

Threshold-Type Edge: Edge as a "Bifurcation Point"

Regime shifts are the most direct case in which Edge is placed at the core of the theory. Ecosystems do not merely change continuously in response to changing conditions; once a threshold, or tipping point, is crossed, they shift abruptly into a different stable state. The sudden transition of a lake from clear water to turbid water is a typical example.

Once the threshold has been crossed, hysteresis, that is, irreversibility, makes return to the original state difficult. This structure gives concrete form to Edge in the five stages, understood

Figure 2: Agriculture and Ecology: A Domain in Which Three Kinds of Edge Converge — ×5

as a state of tension in which opposing elements coexist, as a bifurcation point beyond which one cannot simply return. The precise location of the threshold is difficult to determine in advance, and statistical signs associated with the approach of a critical point, such as rising variance and increasing autocorrelation, have been studied as early warning signals.

Integrated pest management (IPM) is an example in which thresholds have been institutionalized as a management tool. In a decision model where intervention occurs only when pest density exceeds the economic injury level, the threshold functions as a formal bifurcation point that quantitatively defines when intervention should occur. Whereas regime shifts involve thresholds internal to the system, IPM involves thresholds established by human beings. This contrast brings into view the question of whether Edge is something discovered or something designed.

Landscape connectivity and percolation thresholds address thresholds in spatial structure. A slight change in the amount or arrangement of habitat can produce a discontinuous and abrupt change in the success of organism movement or in population response. This framework, which applies percolation theory from physics to ecology, has a dual character: it concerns both connectivity and bifurcation points. The phenomenon in which animals suddenly disappear when forest cover is reduced only slightly offers a vivid demonstration of the nonlinearity near a threshold.

Boundary-Type Edge: Edge as an "Interface"

Ecological boundary theory (ecotones) addresses Edge head-on as the central concept of the theory. In this framework, which generalizes boundaries in terms of patch contrast, the kinds of flows involved, and the properties of the boundary itself, a boundary is treated not as a mere line but as a functional element that transforms flows and organizes interactions.

The claim that "an edge is not a line but a functional element" provides a clue for understanding Edge in the five-stage model not as a static demarcation but as a dynamic apparatus that links and transforms one Field into another. The concept of boundary permeability carries structural implications for other domains as well.

Ecosystem engineering offers the perspective that organisms construct boundary conditions themselves. When beavers build dams and physically rearrange the boundary conditions of a water system, new habitats are created. Whereas other theories analyze an Edge that already exists, this theory addresses the process by which Edge is made. The perspective that Edge is not given but can be constructed dynamically extends the understanding of the five stages.

Relational-Type Edge: Edge as a "Network"

In **plant-soil feedback**, the rhizosphere functions as a nodal point of relation that links plants and soil biota. Plants alter the soil biotic community, and the altered soil then returns either positive or negative effects to the growth of the next generation of plants. In this cyclical framework, Edge appears not as a threshold or interface but as a network of relations.

The fact that different feedback signs, positive or negative, produce different consequences is important. Positive feedback promotes the dominance of particular species, whereas negative feedback promotes species coexistence. This structure, in which the directionality of Vortex determines Bundle, connects directly in agricultural practice to continuous-cropping injury, as a negative feedback effect, and to crop-rotation effects, as the managerial manipulation of feedback direction.

A Visually "Observable" Five-Stage Process: Self-Organization in Drylands

Self-organized patterns in dryland vegetation provide a rare case in which the entire five-stage process is completed within a single phenomenon and can literally be seen in satellite imagery. Under conditions of water limitation, vegetation spontaneously forms regular patterns such as stripes or spots. Self-organization through scale-dependent feedback, combining local facilitation with long-range inhibition, drives the transformation from a homogeneous state, that is, Field, to regular pattern, that is, Bundle.

In this theory, Vortex, in the form of self-organizing feedback, occupies the center, and it is mathematically isomorphic, through Turing instability, with the Belousov-Zhabotinsky reaction in chemistry and with dissipative structures in physics. The recurrent appearance of the same mathematical structure across domains suggests that the structural correspondence of the five stages may not be peculiar to any single domain.

5. Correspondence with the Five Stages

Field: Initial Conditions After Disturbance

In agriculture and ecology, Field appears as the initial conditions prior to the onset of exploration or change: bare ground after disturbance, in ecological succession; the nitrogen pool in soil, in nitrogen cycling; the mosaic structure of a landscape, in percolation; and homogeneous environmental conditions, in dryland vegetation. In many cases, this is either the state in which the Bundle of the previous cycle has collapsed or a state in which new conditions have been established.

Wave: The Manifestation of Fluctuation and Difference

The invasion of pioneer species and competition over resources, in ecological succession; the biasing of nitrogen forms by temperature and moisture variation, in nitrogen cycling; increases and decreases in pest populations, in IPM; and the destabilization of systems by shifting environmental conditions, in regime shifts: all of these are stages in which differences and fluctuations become manifest out of the initial conditions.

Edge: Diverse Implementations in Three Forms

As described in detail in the previous section, three different types of Edge were confirmed in agriculture and ecology. Threshold-type Edge implements Edge as a bifurcation point, boundary-type Edge implements it as an interface, and relational-type Edge implements it as a network. This diversity suggests that Edge in the five-stage model does not refer to a single phenomenon but is a concept that encompasses multiple structural patterns.

Vortex: The Formation of Order Through Feedback

Feedback, that is, Vortex, was explicitly confirmed in all 10 cases. In particular, three cases place Vortex at the very core of the theory: self-reinforcing feedback in regime shifts, scale-dependent feedback in dryland vegetation, and positive and negative feedback in plant-soil interaction. In IPM and state-and-transition models, Vortex appears as a management cycle of intervention, ecosystem response, and reevaluation.

Bundle: Stable States and the Institutionalization of Management

Climax community structure, in ecological succession; yield levels and residual nitrogen amounts, in nitrogen cycling; regionally standardized IPM protocols, in IPM; new stable states, in regime shifts; and regular spatial patterns, in dryland vegetation: Bundle becomes fixed as a stable

structure. In agriculture, Bundle is often institutionalized in the form of management guidance or guidelines. The example of the USDA adopting state-and-transition models as rangeland management guidance succinctly demonstrates the institutionalization of Bundle.

6. Insights from This Domain

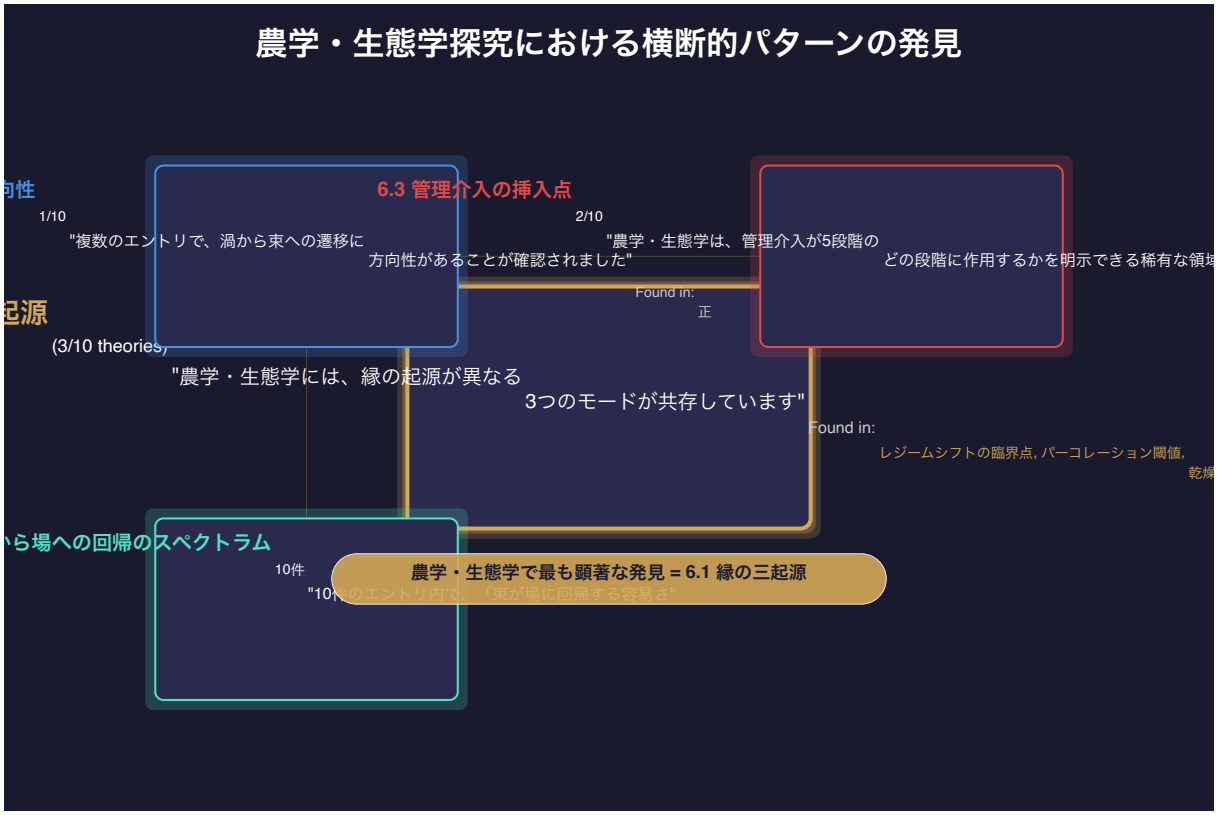

Figure 3: Agriculture and Ecology: A Domain in Which Three Kinds of Edge Converge —

Three Types of Edge

The most important insight that agriculture and ecology contribute to the investigation as a whole is the finding that Edge can be implemented in at least three different forms.

- **Threshold type**: A bifurcation point at which a system, once certain conditions are exceeded, shifts into a qualitatively different state (regime shifts, percolation thresholds, the economic threshold in IPM, transition conditions in state-and-transition models)
- **Boundary type**: An interface that transforms flows and organizes interactions between different systems (ecotones, the material interface of the rhizosphere, the structures built by ecosystem engineers)
- **Relational type**: A nodal point in a network that links multiple elements (interspecies interactions, host specificity in plant-soil feedback)

This classification provides a framework that may also be applicable to other domains.

"Discovered" Edge and "Designed" Edge

The contrast between IPM, where thresholds are established by human beings, and regime shifts, where thresholds are internal to the system, together with the findings of ecosystem engineering, where organisms construct boundaries, brings into view the question of whether Edge is something discovered or something designed. This question remains for continued examination.

Fractality Across Scales

From the molecular interface of the rhizosphere, in nitrogen cycling, to interspecies relations at the population level, in ecological succession; from patch-to-patch connection at the landscape level, in percolation, to regime shifts at regional and planetary scales: a tendency was confirmed for the structure of relation to repeat the same pattern across different scales.

7. Unresolved Questions

- **Integration of the three types of Edge**: Are threshold type, boundary type, and relational type independent concepts, or are they different aspects of a single structure? No conclusion has yet been reached on this question.
- **The scope of "designed Edge"**: The findings of ecosystem engineering and IPM suggest that Edge can be constructed. However, where is the boundary between constructible Edge and non-constructible Edge?
- **The relation between management intervention and the five stages**: At which stage of the five stages does agricultural management act? Under what conditions does management intervention promote the circulation of the five stages, and under what conditions does it inhibit it?
- **Discontinuous change and the five stages**: How should the discontinuous changes shown by regime shifts be reconciled with the cyclical model of the five stages? This tension has been recognized but not resolved.

8. Conclusion

Agriculture and ecology are domains that show very high structural correspondence with the five-stage model. All 10 candidates were judged adoptable, and 5 of them received the highest-intensity evaluation in their correspondence with Edge.

The most important contribution of this domain is that it shows Edge can be implemented in three different forms: threshold type, boundary type, and relational type. This diversity suggests that Edge in the five-stage model is not a single phenomenon but a concept that encompasses multiple structural patterns.

Some of these correspondences, however, involve interpretation. In particular, Edge in ecological succession and nitrogen cycling does not possess a structure as explicit as that found in regime shifts or boundary theory. Moreover, the relation between discontinuous change and the circulation of the five stages remains as an unresolved question.

This investigation confirmed that agriculture and ecology provide a rich basis for understanding the structure of the five-stage model, especially the diversity of Edge. At the same time, it also reports that there are clear differences in the strength of correspondence and that not every case holds with the same degree of certainty.